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> PROJECT OVERVIEW · NOVEMBER 2026

ZÆTHERION

A desktop for my laptop, designed from scratch.

What I'm building, why, and how I'm doing it carefully.

zaetherion-shell · design stage · no code yet

> WHAT IS THIS

The part of the computer you actually look at

When you use a phone, there is the home screen, the bar at the top with the time and battery, the notifications that slide down, the lock screen. None of that is an app: it is the **shell**, the frame around the apps. Computers have one too.

My laptop runs Linux, where you can choose, and even build, your own shell. For the past weeks I have been building a control panel called **FART** (a codename that stuck): one panel at the top of the screen with music, timers, tasks, notifications and system controls, all in my own visual style. It runs next to a ready-made shell that still handles everything else.

ZÆTHERION is the next step: turning that panel into the *whole* shell, the bar, the app launcher, notifications, the lock screen, the wallpaper, so the entire screen speaks one design language that I designed.

Why build it?

Partly because I use it every day and want it to be exactly right. Partly because it is a serious engineering exercise. It touches interface design, performance measurement, system programming, security (the lock screen) and open-source licensing, and it has to be reliable enough to trust with my own work and study.

Where it stands today

It is fully **designed but not yet built**. I spent a long session answering every design question in advance, from how the clock looks to what happens if part of the system crashes. The result is 142 written decisions, so nothing has to be guessed during building. Building starts once the current panel project reaches its 12th stage.

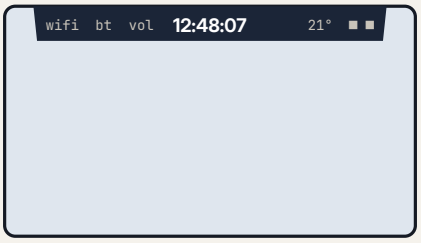
> THE CENTREPIECE

The island

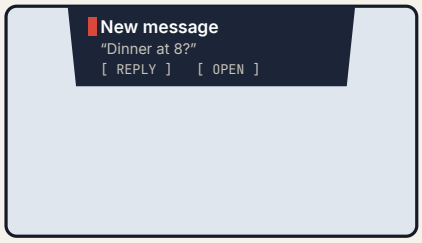
Instead of a full bar across the top of the screen, there is one small dark shape hugging the top edge, like the notch on a phone. Its colour is the dark navy of the laptop's own body. Everything else grows out of it and shrinks back into it.



1 At rest



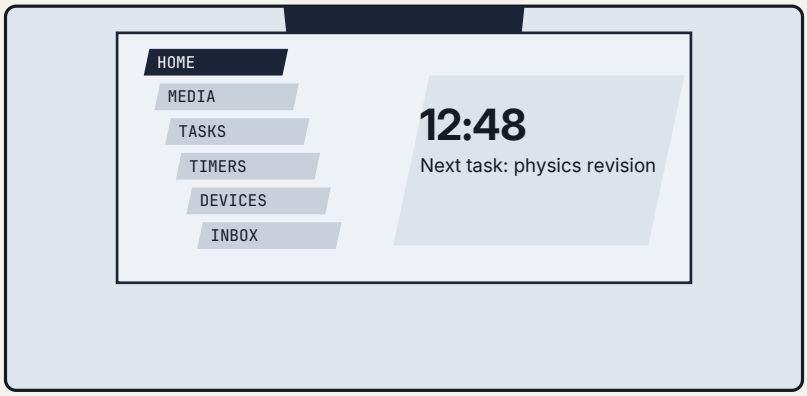
2 Mouse hovers



3 Something happens

At rest it shows only the time, the song playing (a tiny cover and a thin moving line), and the battery. **When the mouse hovers**, it spreads out sideways like wings to show Wi-Fi, Bluetooth, volume, the weather and small app icons. **When something happens** (a message, a finished timer, the volume changing, a low battery) it grows to show it, then shrinks back after a few seconds.

If several things happen at once, a fixed order of importance decides: an urgent alert beats a timer, which beats a song change, and so on.



Clicking the island unfolds the full control panel out of it: rooms for music, tasks, timers, devices, notifications and more. Clicking again folds it back.

> THE LOOK

A real sky, and a bold graphic style

The wallpaper and the lock screen can show the **actual sky above where I am**, worked out from astronomy formulas: the stars, the Moon with its current phase, the planets that are up, and the Sun's height during the day. At night it is drawn like an astronomer's star chart; in the day it becomes a soft gradient that follows the Sun.



The graphic style borrows *ideas* from two video games I like, never their actual pictures, fonts or names:

Persona	bold slanted shapes, everything leaning at the same angle, events that ‘cut in’ like a comic panel
NieR: Automata	the computer speaks in short machine-like lines, such as `> WIFI · ONLINE`; this can be switched off
Astronomy	the real sky, orbits as motion, light bending around mass as a transition effect

The lock screen becomes a ‘title card’: a large slanted clock and the date over the sky, the password box underneath, and music controls and the battery along the bottom.

> USING LITTLE

Beautiful, but light on the laptop

Pretty desktops often waste battery and memory on animations nobody is looking at. ZÆTHERION has two modes, and the rule is simple: **nothing moves unless I ask it to**.



Silent is the everyday mode. The screen is still; the clock changes once a minute; the sky is redrawn at most once a minute. When nothing is happening, the shell does essentially no work at all.

Overdrive is switched on deliberately, by holding a key or pressing and holding the island, for a set time, for example twenty minutes. Then the sky slowly turns, song lyrics animate word by word, and transitions get cinematic effects. When the time is up, everything returns to Silent and the memory it used is given back. It switches itself off on battery, and it never wakes the laptop's power-hungry gaming graphics card.

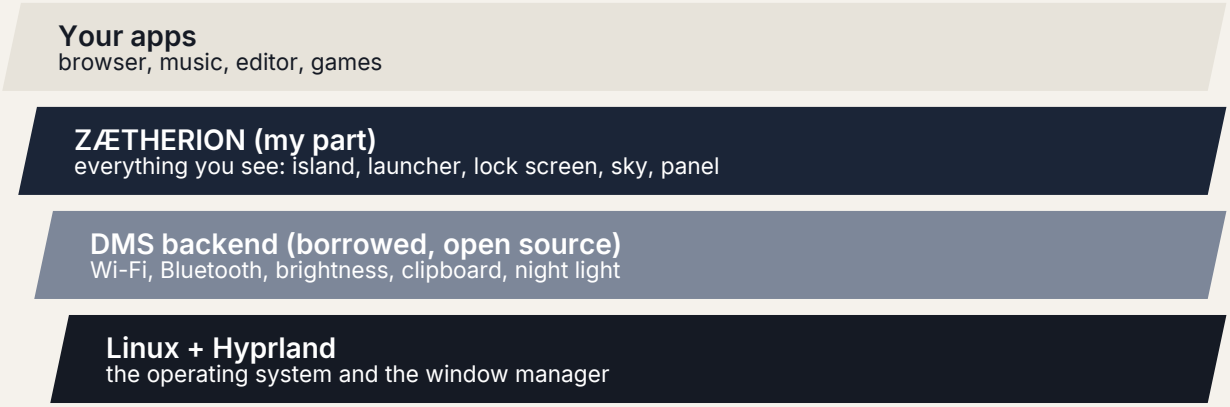
Other habits that keep it light

• Anything not visible is removed from memory, not just hidden. Desktop widgets disappear completely whenever an app window is open.
• Every part has a measured budget, and the whole shell must use no more memory than the setup I use today.
• Nothing is called ‘faster’ or ‘lighter’ until it has been measured.

> BUILT CAREFULLY

How I'm making it safe to rely on

A shell is something you trust all day: if the lock screen fails, the computer is open; if notifications break, you miss messages. So the plan is cautious by design.



I don't reinvent the plumbing. Connecting to Wi-Fi, pairing Bluetooth, controlling brightness and the clipboard are handled by **DMS**, a well-tested open-source project. I build the part you see and interact with; proven code does the rest.

The old desktop stays as a safety net. ZÆTHERION runs as a separate choice on the login screen. If anything goes wrong, I log out and pick the old one. It is only retired after a full test week with no problems in the last three days.

Security parts are tested separately. The lock screen and password prompts are tested inside a sandboxed copy of the desktop first, never by risking the real one.

Everything is written down. Every decision, every risk with its fix, and every feature of the old desktop (kept, replaced or dropped) is recorded in the project's public documents, so nothing is left to chance or memory.

> THE PLAN

Step by step

The work is split into thirteen stages, S0 to S12. Each one ends with something that works, checked against written pass/fail criteria, before the next begins.



What I learn along the way

Design	a complete, consistent visual system: shapes, type, colour, motion
Engineering	several programs working together, recovering on their own when one fails
Performance	measuring memory, processor time and smoothness instead of guessing
Security	a lock screen and password prompts that fail safely
Open source	reusing others' work properly: licences, credit, publishing my own
Planning	142 decisions made up front, risks listed with their fixes

The whole plan is public on GitHub as `zaetherion-shell`. For now it holds the design documents; the code arrives stage by stage.